Annotating the Functional Roles of Social-Person References in Bipolar-Related Suicidal Ideation Narratives

Salim Sazzed
Georgia Southern University
Statesboro, Georgia, USA
ssazzed@georgiasouthern.edu

Farhan Noor Dehan
Georgia Southern University
Statesboro, Georgia, USA

Pronoy Sarker
Georgia Southern University
Statesboro, Georgia, USA

Abstract

Online suicidal ideation posts in Reddit's *r/SuicideWatch* community often include references to close others, such as family members, romantic partners, ex-partners, friends, and loved ones. These references cannot be fully understood by relationship category alone because the same category may appear in different narrative contexts. In this paper, we present a reference-level annotation framework for identifying the roles of real-life social-person references in bipolar-related suicidal ideation posts. Applying the framework to 300 randomly sampled bipolar-related *r/SuicideWatch* posts produces 1,072 valid social-person reference annotations. References to family members are the most common person-type category (53.45%), followed by friend/peer (22.76%) and romantic partner/ex-partner (19.03%). Disconnection/loss/rejection is the most frequent role label (28.26%), followed by conflict/harm/betrayal (23.88%), background/unclear (19.87%), protective/supportive (16.88%), and burden/guilt toward others (11.10%). Person-type-by-role analysis shows that family references are most often associated with conflict/harm/betrayal, while friend/peer references are most often associated with disconnection/loss/rejection. Each annotation is grounded in an explicit evidence span to reduce unsupported interpretation. The framework supports structured and interpretable analysis of interpersonal meaning in suicidal ideation narratives.

Keywords

mental health AI, suicidal ideation, bipolar disorder, social relationships, functional role

1 Introduction

Online mental health communities contain rich first-person accounts of psychological distress, suicidal ideation, and the social circumstances surrounding these experiences [1, 3, 6, 7].In suicidal ideation narratives, authors often describe their distress in relation to people who are personally meaningful to them, including parents, siblings, romantic partners, ex-partners, friends, and loved ones [4, 8, 9]. These interpersonal references can provide important context for understanding the narrative, but their meaning depends on how the referenced person is positioned in the surrounding text.

This distinction is important because the same person category can appear in different narrative roles. A family member may be described as a reason the author remains alive, someone the author does not want to burden, someone who has died or become unavailable, or someone involved in conflict or mistreatment. Similarly, a romantic partner, ex-partner, or friend may appear in contexts of support, breakup, rejection, abandonment, betrayal, or concern about causing worry. Treating these cases only as person-category mentions removes the functional meaning of the reference.

Bipolar-related suicidal ideation posts provide a meaningful setting for this task because bipolar disorder is associated with elevated suicide risk, and prior work has linked suicidal ideation in bipolar disorder to interpersonal factors such as family and peer relationship quality, social support, and family functioning [2, 4, 8]. In online suicidal ideation narratives, references to close others may therefore appear in different interpersonal contexts, including support, burden, rejection, loss, or conflict. This makes it important to distinguish not only which person is mentioned, but how that person functions in the surrounding narrative.

In this paper, we present a reference-level annotation framework for identifying the functional roles of real-life social-person references in bipolar-related suicidal ideation posts. Reference-level annotation is necessary because a single post may mention multiple people, and each reference may serve a different function in the narrative. Each annotation includes the reference text, an evidence span, a person-type label, and a role label. This design captures both the relationship category of the referenced person and the functional role that person plays in the suicidal narrative.

The framework uses four person-type labels: family, romantic partner or ex-partner, friend or peer, and close unspecified person. It then assigns one of five functional-role labels: protective/supportive, burden or guilt toward others, disconnection/loss/rejection, conflict/harm/betrayal, and background/unclear. Together, these labels separate the relationship category from the narrative function of the referenced person.

The framework is designed to keep annotations grounded in explicit textual evidence. Annotators are instructed to assign labels only when the post provides direct evidence for the relationship type and functional role of the referenced person. This evidence-grounded design supports responsible analysis of sensitive mental health narratives by identifying interpersonal roles without turning the annotation task into clinical interpretation or suicide-risk assessment.

This paper makes two contributions. First, it presents an annotation framework for social-person references in bipolar-related suicidal ideation discourse, including person-type categories, role labels, inclusion and exclusion criteria, evidence-span requirements, and decision rules so that each role label is grounded in explicit text from the post. Second, it applies the framework to annotate a subset of bipolar-related *r/SuicideWatch* posts and reports person-type, role-label, and person-type-by-role distributions, showing that different relationship categories are associated with distinct patterns of social-person roles in these narratives.

2 Data and Annotation Framework

2.1 Corpus Creation

We analyze posts from a publicly available corpus of 27,334 Reddit suicidal discourse posts[1]. Reddit has been widely used to study mental health self-disclosure, social support, and suicidal ideation because its topic-based communities allow users to discuss sensitive experiences with relative anonymity [1, 5, 10]. Specifically, *r/SuicideWatch* is a peer-support community where users discuss suicidal thoughts, psychological distress, and related life experiences.

From this corpus, we identified posts that explicitly referenced bipolar disorder using a keyword-based filtering strategy. Posts were selected if they contained the terms *bipolar* or *bipolar disorder*, resulting in 1,175 posts. This filtering step was used to construct a bipolar-related subset based on explicit textual references.

As a preliminary study, we randomly sampled 300 posts from the 1,175 bipolar-related posts for reference-level annotation. This annotated subset was used to develop and evaluate the proposed annotation framework.

Table 1: Annotation fields for reference-level social-person role labeling.

Field	Description
post_id	Identifier for the post
reference_id	Identifier for the social-person reference
reference_text	Exact person or group phrase
evidence_text	Text span supporting the role label
person_type	Family, romantic partner/ex, friend/peer, or close unspecified person
role	Functional role label
notes	Optional explanation for difficult cases

2.2 Annotation Unit

The annotation unit is the reference-level social-person mention, not the entire post. A social-person reference is defined as a specific mention of a real-life or close interpersonal person or group known to the user. Examples include family members, romantic partners or ex-partners, friends, roommates, classmates, coworkers, and personally meaningful loved ones.

For example, in the sentence, 'The only reason I have not killed myself is because of my mom and brothers. After my dad passed away, I fell into a deep dark hole," there are two separate social-person references. The phrase 'my mom and brothers" functions as a protective or supportive reference, while "my dad" functions as a loss-related reference. These should be annotated as separate rows rather than collapsed into one post-level label.

This reference-level design allows the framework to capture multiple interpersonal functions within the same narrative. It also enables future models to identify specific evidence spans and person-role associations.

[1]https://github.com/sazzadcsedu/BipolarDisorder.git

2.3 Criteria for Social-Person References

This task focuses on reference-level mentions of real-life social-person relationships in bipolar-related suicidal ideation posts. A social-person reference is included when the text names or clearly refers to a real-life person or close interpersonal group known to the author. These references are annotated separately, even when multiple references appear in the same post.

Family includes family members or family groups, such as mother, father, parents, siblings, brother, sister, children, cousins, or family.

Romantic partner or ex-partner includes current or former romantic partners, such as girlfriend, boyfriend, wife, husband, partner, spouse, or ex-partner.

Friend or peer includes friends and real-life peer contacts, such as friends, best friends, roommates, coworkers, classmates, or peers.

Close unspecified person is used only when the text clearly refers to close or personally meaningful people but does not specify the exact relationship. Examples include loved ones, people I care about, the ones I love, or people who care about me.

References are excluded when they do not refer to close real-life social-person relationships. Excluded references include care professionals such as therapists, psychiatrists, doctors, and counselors; online audiences such as Reddit users, commenters, online strangers, or people online; vague or generic groups such as people, everyone, others, society, no one, or nobody; pets; institutions; and mental health conditions.

Some absence statements are included when they name a specific social-person group. For example, "I have no friends" is annotated as a friend/peer reference, and "I have no family" is annotated as a family reference. In contrast, statements such as "No one cares about me" or "People always leave me" are excluded unless the surrounding text clearly identifies a specific close social-person group.

2.4 Social-Person Role Labels

Each identified social-person reference is assigned one role label based on explicit textual evidence. When the same person is clearly associated with more than one function in the post, annotators create separate annotation rows, each linked to the evidence supporting that role. The framework defines five role labels.

2.4.1 Protective or Supportive. The *protective or supportive* label is assigned when the referenced person or group is explicitly described as a source of support, a reason for continued survival, a reason for not acting on suicidal thoughts, someone who provided help or rescue, or someone the author contacted for help.

This label requires explicit evidence of support, help, rescue, or reason-to-live meaning. A family member, friend, or partner should not be coded as protective/supportive solely because they are mentioned in the post.

2.4.2 Burden or Guilt Toward Others. The *burden/guilt toward others* label is assigned when the author explicitly expresses concern about hurting, burdening, worrying, frightening, disappointing, or causing grief to a close person or group.

This label captures cases in which the author frames their suffering, suicidal thoughts, or possible death in terms of its emotional

Table 2: Examples of reference-level social-person role annotations.

Reference	Evidence span	Person type	Functional role
girlfriend	"my long distance girlfriend who had stuck with me through all this"	Romantic partner/ex	Protective/supportive
family	"i dont want to be a burden on my family"	Family	Burden/guilt toward others
parents	"my parents had stopped talking to me since my recent diagnosis of bipolar disorder"	Family	Disconnection/loss/rejection
dad	"my dad... burst into my room while i was sleeping and yelled"	Family	Conflict/harm/betrayal
friends	"i walk into the office, i make jokes, i talk to friends"	Friend/peer	Background/unclear

consequences for others. The label is used only when the post explicitly expresses guilt, burden, worry, fear of causing distress, or concern about the impact on others.

2.4.3 Disconnection, Loss, or Rejection. The *disconnection, loss, or rejection* label is assigned when the referenced person is described as absent, unavailable, lost, rejecting, or disconnected from the author. This includes abandonment, lack of friends or family, stopped communication, breakup, death or bereavement, lack of support, being ignored, or not being listened to.

Breakups, bereavement, abandonment, and being cut off are generally coded under this label unless the text explicitly describes abuse, cheating, violence, betrayal, or another form of direct interpersonal harm.

2.4.4 Conflict, Harm, or Betrayal. The *conflict/harm/betrayal* label is assigned when the referenced person or group is explicitly associated with interpersonal conflict, mistreatment, abuse, betrayal, violence, or harmful interaction. This includes cheating, repeated fighting, yelling, physical violence, confinement, coercive behavior, or abusive family or partner dynamics.

This label requires explicit evidence of conflict, harm, betrayal, abuse, violence, or mistreatment. It should not be assigned solely because the author is distressed about the person or relationship.

2.4.5 Background or Unclear. The *background or unclear* label is assigned when a real-life social person is mentioned, but the post does not provide sufficient evidence to determine the person's functional role in the suicidal narrative.

Examples include statements such as 'I live with my parents," 'I talk to friends and family," or "my brother is 25," when no additional context indicates support, burden, disconnection, rejection, conflict, harm, or betrayal. This label functions as a fallback category and should not be used when one of the four substantive role labels is clearly supported by the text.

2.5 Annotation Procedure and Decision Rules

The annotation codebook is publicly available in the GitHub repository[2]. Each annotation row corresponds to one eligible social-person reference. The row records the post identifier, reference identifier, exact reference text, supporting evidence span, person-type label, role label, and optional notes. The evidence span is required so that each label is grounded in explicit language from the post. The notes field is used only for difficult or ambiguous cases. The annotation fields are summarized in Table 1.

Table 3: Distribution of annotated social-person references by person type and role. Percentages are computed over 1,072 valid annotations.

Label type	Label	Count	Percentage
Person type	Family	573	53.45%
	Friend/peer	244	22.76%
	Romantic partner/ex	204	19.03%
	Close unspecified person	51	4.76%
Role	Disconnection/loss/rejection	303	28.26%
	Conflict/harm/betrayal	256	23.88%
	Background/unclear	213	19.87%
	Protective/supportive	181	16.88%
	Burden/guilt toward others	119	11.10%

Annotators follow five decision rules. First, labels must be based only on explicit textual evidence. Annotators should not infer hidden meaning, relationship quality, clinical severity, or suicide risk beyond what is written. For example, "I live with my parents" is labeled as background/unclear unless additional text explicitly indicates support, burden, disconnection, rejection, conflict, or harm.

Second, each eligible social-person reference is annotated separately. If a post states, "My family stopped talking to me, but my friend still tries to help," the family reference is labeled as disconnection/loss/rejection, while the friend reference is labeled as protective/supportive.

Third, the same person may receive more than one role only when the text clearly provides evidence for multiple functions. In such cases, annotators create separate rows for the same reference, each linked to the evidence supporting that role. For example, if a post states, 'My mom is the only reason I am alive, but I feel like a burden to her," annotators create two rows for 'mom": one protective/supportive and one burden/guilt toward others.

[2]https://github.com/sazzadcsedu/BipolarDisorder.git

Table 4: Distribution of functional-role labels within each person-type category. Percentages are computed within each person type.

Person Type	Disconnection/ Loss/Rejection	Conflict/ Harm/Betrayal	Background/ Unclear	Protective/ Supportive	Burden/ Guilt
Family (n = 573)	125 (21.8%)	161 (28.1%)	133 (23.2%)	92 (16.1%)	62 (10.8%)
Friend/Peer (n = 244)	104 (42.6%)	34 (13.9%)	43 (17.6%)	41 (16.8%)	22 (9.0%)
Romantic Partner/Ex (n = 204)	63 (30.9%)	50 (24.5%)	25 (12.3%)	41 (20.1%)	25 (12.3%)
Close Unspecified (n = 51)	11 (21.6%)	11 (21.6%)	12 (23.5%)	7 (13.7%)	10 (19.6%)

Fourth, loss or separation is distinguished from betrayal or harm. For example, 'My girlfriend left me" is coded as disconnection/loss/rejection, while 'My girlfriend cheated on me" is coded as conflict/harm/betrayal.

Fifth, death and bereavement are coded as disconnection/loss/rejection unless another role is explicitly stated. References to a parent, friend, partner, or other close person who died are treated as loss-related when the post connects that death to absence, grief, or distress.

We applied the annotation framework to 300 bipolar-related suicidal ideation posts. Each post was screened for eligible real-life social-person references, and each eligible reference was annotated with a person-type label and a functional-role label. Because the task is reference-level, the final number of annotations exceeds the number of posts.

Annotator 1 completed the initial annotations, and Annotator 2 subsequently reviewed the existing annotations for missed references, incorrect labels, inconsistent decisions, and cases requiring clarification. Questionable items were highlighted in the shared sheet and re-checked against the full source context and annotation guideline. Cases involving unclear evidence, overlapping possible roles, ambiguous relationship references, or inconsistent treatment of similar examples were discussed and resolved through consensus when possible. Annotator 2 did not create a separate independent annotation set; therefore, formal inter-annotator agreement analysis was not conducted for this preliminary work. Future work will include independent double annotation and formal inter-annotator agreement analysis as the annotated corpus is expanded.

3 Annotation Results

The annotation process produced 1,072 valid social-person reference annotations. Table 3 summarizes the distributions of person-type and role labels.

Family references were the most frequent person type, accounting for 53.45% of valid annotations. Friend/peer references accounted for 22.76%, followed by romantic partner/ex-partner references at 19.03% and close unspecified persons at 4.76%.

For functional roles, disconnection/loss/rejection was the most frequent category, accounting for 28.26% of valid annotations. Conflict/harm/betrayal accounted for 23.88%, followed by background/unclear at 19.87%, protective/supportive at 16.88%, and burden/guilt toward others at 11.10%. These distributions suggest that interpersonal references in bipolar-related suicidal ideation posts are functionally heterogeneous, with references appearing in contexts of loss, conflict, support, burden, and unclear background description.

To further examine how functional roles varied across person-type categories, Table 4 reports the role distribution within each person type.

Family references were most often associated with conflict, harm, or betrayal, whereas friend/peer references were most often associated with disconnection, loss, or rejection. Romantic partner/ex-partner references were distributed mainly across disconnection, loss, or rejection, conflict/harm/betrayal, and protective/supportive roles. These patterns further show that person-type categories alone do not capture the interpersonal function of social-person references.

4 Discussion

Annotation results indicate that social-person references in bipolar-related suicidal ideation posts are functionally diverse rather than confined to a single interpersonal role. Family was the most frequent person type, suggesting that family relationships were prominent in this annotated subset. However, the role distribution shows that social-person references should not be interpreted as uniformly supportive, harmful, or peripheral. Instead, they appeared across contexts of disconnection, conflict, background mention, support, and burden-related concern.

The person-type-by-role distribution further shows that the meaning of a social-person reference varies by relationship category. Family references were most often associated with conflict/harm/betrayal, while friend/peer references were most often associated with disconnection/loss/rejection. Romantic partner/ex-partner references were distributed across disconnection/loss/rejection, conflict/harm/betrayal, and protective/supportive roles. These patterns suggest that person-type labels alone do not provide enough information about how close others are positioned in suicidal ideation narratives.

More broadly, these findings highlight the value of treating interpersonal context as a structured component of suicidal ideation narratives rather than as incidental background information. Social-person references can indicate how authors situate distress within relationships, including experiences of support, loss, conflict, rejection, and concern for others. Representing these distinctions makes computational analysis more interpretable by linking each role label to a specific referenced person and its local narrative evidence.

The evidence-grounded design also has implications for mental-health NLP and large language model evaluation. Because each annotation is linked to a specific reference and supporting evidence

span, the labels can be used to examine whether models recognize interpersonal roles from the text rather than relying on broad assumptions about family members, partners, or friends. This is especially important for sensitive mental-health narratives, where the same relationship category may carry different meanings depending on the local context.

5 Limitations and Future Work

This study has several limitations. First, the framework depends on explicit textual evidence. This design reduces unsupported inference, but it may miss interpersonal meanings that are implied rather than directly stated. Second, the framework focuses on close real-life social-person references and does not cover other socially relevant entities, such as online audiences, care professionals, institutions, pets, or vague collective references. Third, the current analysis is based on bipolar-related suicidal ideation posts identified through explicit keyword mentions, so the findings should be interpreted as applying to this annotated subset rather than to all posts authored by individuals with bipolar disorder.

Future work will extend the annotated corpus, measure inter-annotator agreement, and refine label boundaries for ambiguous cases. We also plan to conduct post-level co-occurrence analyses to examine how different person types and functional roles appear together within the same post. This would allow us to study whether certain relationship categories are commonly associated with particular roles, such as whether family references frequently co-occur with burden-related or conflict-related roles, and whether multiple interpersonal functions commonly appear in the same narrative. Future work will also develop baseline models for automatic reference-level person-type and functional-role classification and examine whether large language models can follow evidence-grounded annotation rules without over-inferring interpersonal meaning in sensitive mental health narratives.

6 Summary and Conclusion

We presented a reference-level annotation framework for identifying the roles of real-life social-person references in bipolar-related suicidal ideation narratives from Reddit's *r/SuicideWatch*. For each annotated reference, the framework records the referenced person or group, the supporting evidence span, the relationship category, and the role of that reference in the surrounding narrative. The results show that social-person references are functionally diverse, spanning support, burden or guilt, disconnection or loss, conflict or harm, and background context.

These findings demonstrate that relationship categories alone are insufficient for capturing how close others are positioned in suicidal ideation narratives. By grounding each annotation in explicit textual evidence, the framework provides a structured basis for analyzing interpersonal meaning in sensitive mental health discourse. More broadly, this work can support the development of interpretable datasets and future evaluation of NLP and large language model systems that analyze interpersonal context in mental health narratives.

References

[1] Munmun De Choudhury and Sushovan De. 2014. Mental health discourse on reddit: Self-disclosure, social support, and anonymity. In *Proceedings of the international AAAI conference on web and social media*, Vol. 8. 71–80.

[2] Wenwen Dou, Lu Zhang, Yaping Zhang, Hong Li, Yiru Wang, Rui Chen, and Gang Wang. 2022. Family and psychosocial functioning in bipolar disorder. *Frontiers in Psychology* 12 (2022), 807546.

[3] Manas Gaur, Ugur Kursuncu, Amanuel Alambo, Amit Sheth, Raminta Daniulaityte, Krishnaprasad Thirunarayan, and Jyotishman Pathak. 2018. " Let Me Tell You About Your Mental Health!" Contextualized Classification of Reddit Posts to DSM-5 for Web-based Intervention. In *Proceedings of the 27th ACM international conference on information and knowledge management*. 753–762.

[4] Shari Greenberg, Kathryn L. Rosenblum, Melvin G. McInnis, and Maria Muzik. 2014. The role of social relationships in bipolar disorder: A review. *Psychiatry Research* 219, 2 (2014), 248–254.

[5] Sagar Joglekar, Sumithra Velupillai, Rina Dutta, and Nishanth Sastry. 2021. Analysing Meso and Macro Conversation Structures in an Online Suicide Support Forum. *BMC Psychiatry* 21 (2021), 1–17.

[6] Virginia Morini, Maria Sansoni, Giulio Rossetti, Dino Pedreschi, and Carlos Castillo. 2025. Participant behavior and community response in online mental health communities: Insights from Reddit. *Computers in Human Behavior* 165 (2025), 108544.

[7] Umashanthi Pavalanathan and Munmun De Choudhury. 2015. Identity management and mental health discourse in social media. In *Proceedings of the 24th international conference on world wide web*. 315–321.

[8] Craig J. R. Sewall, Tina R. Goldstein, Aidan G. C. Wright, Daniel Rosen, Neal D. Ryan, Shirley Yen, Heather Hower, Fangzi Liao, and Boris Birmaher. 2020. Interpersonal relationships and suicidal ideation in youth with bipolar disorder. *Archives of Suicide Research* 24, 2 (2020), 236–249.

[9] Kimberly A. Van Orden, Tracy K. Witte, Kelly C. Cukrowicz, Scott R. Braithwaite, Edward A. Selby, and Thomas E. Joiner. 2010. The Interpersonal Theory of Suicide. *Psychological Review* 117, 2 (2010), 575–600.

[10] Eldar Yeskuatov, Sook-Ling Chua, and Lee Kien Foo. 2022. Leveraging reddit for suicidal ideation detection: a review of machine learning and natural language processing techniques. *International journal of environmental research and public health* 19, 16 (2022), 10347.